CHAPTER 9

Joints

How bones connect — and how the body moves.

Anatomy & Physiology · Broward-Miami Health Institute

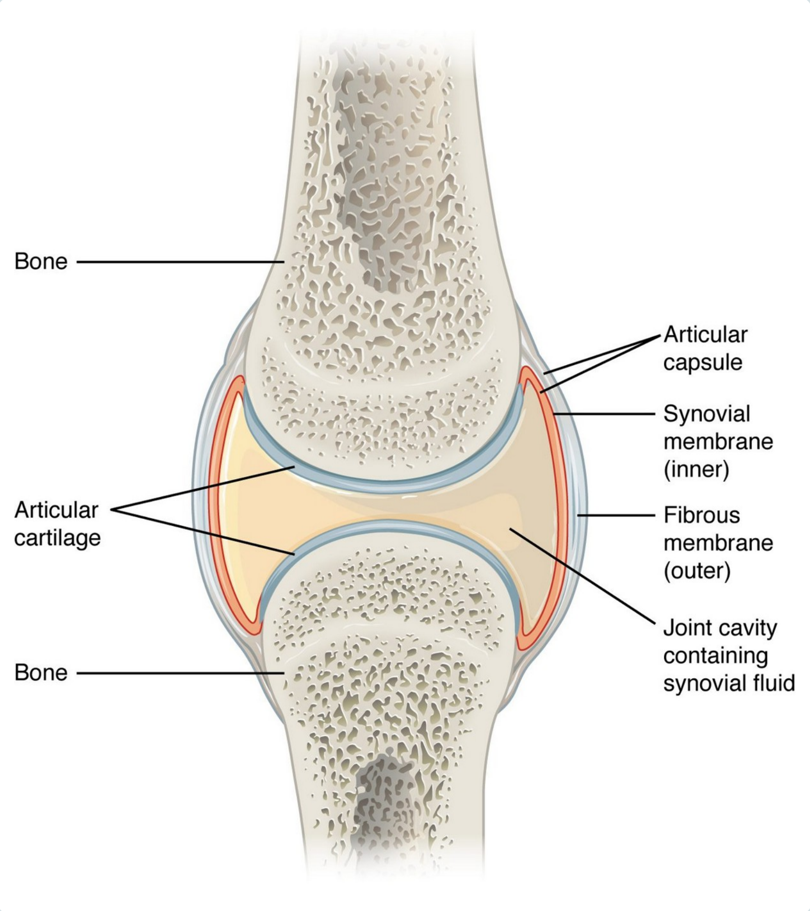
FACULTY EDITION

CHAPTER 9

Learning Objectives

After studying this chapter, students will be able to:

1. Discuss both functional and structural classifications for body joints
2. Describe the characteristic features for fibrous, cartilaginous, and synovial joints and give examples of each
3. Define and identify the different body movements
4. Discuss the structure of specific body joints and the movements allowed by each



A synovial joint — the body's most movable joint.

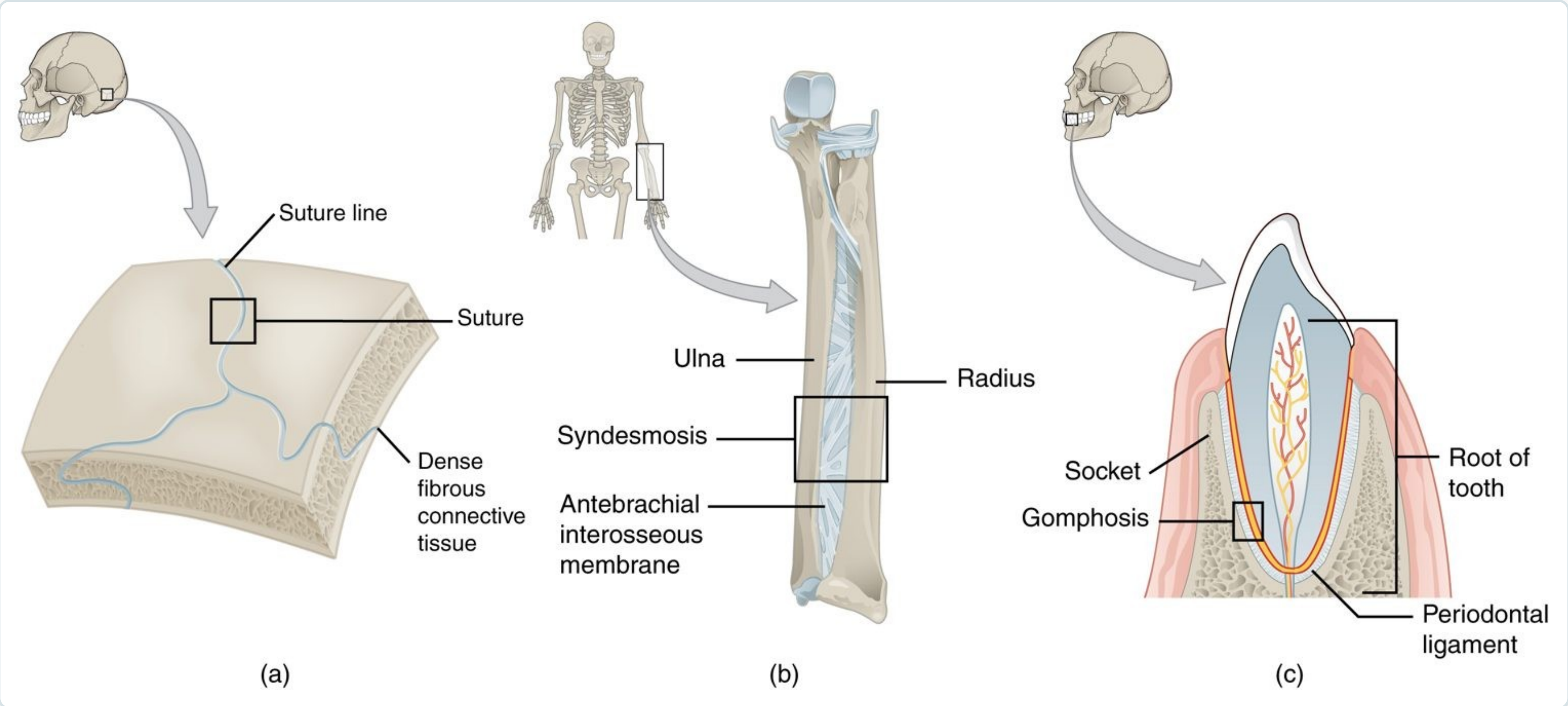
9.2

SECTION

Fibrous Joints

Joints held by fibrous connective tissue — mostly immovable.

Fibrous Joints



Types of fibrous joints

- Bones joined by fibrous tissue
- Little or no movement
- Sutures join the skull bones
- Gomphoses anchor the teeth

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SECTION

Cartilaginous Joints

Joints united by cartilage — slightly movable or immovable.

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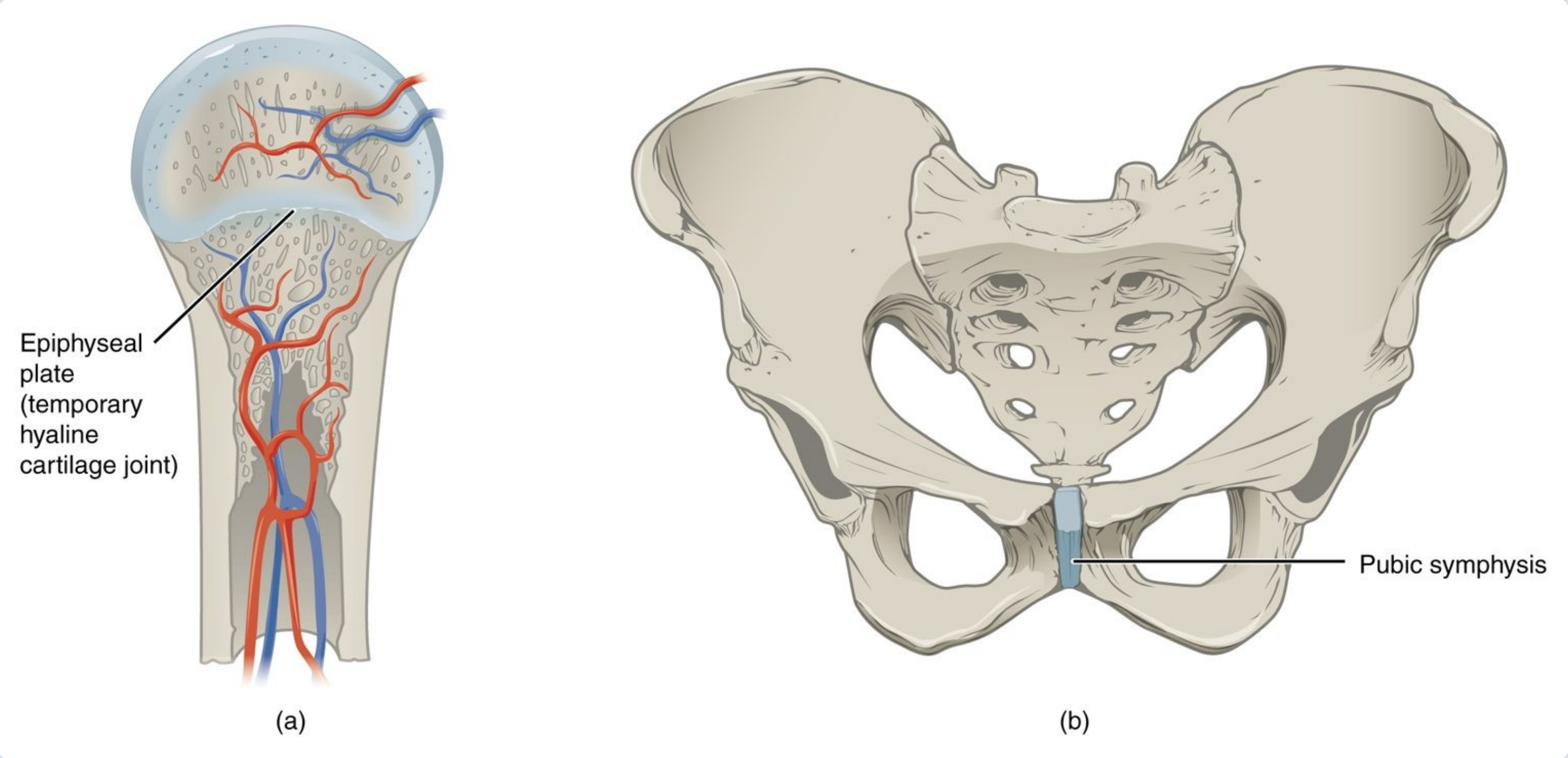
Cartilaginous Joints



Types of cartilaginous joints

- Bones joined by cartilage
- Slightly movable or immovable
- Pubic symphysis allows give in birth
- Intervertebral discs cushion the spine

9.4

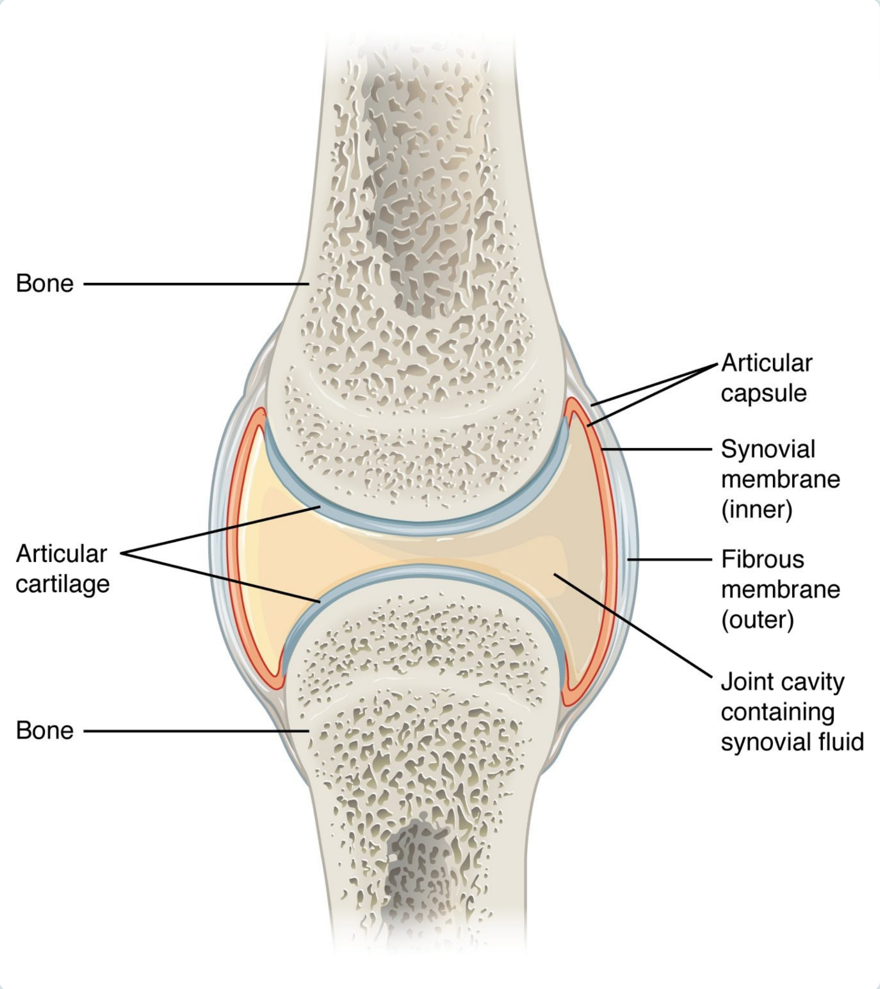
SECTION

Synovial Joints

Freely movable joints with a fluid-filled cavity — the most common type.

SECTION 9.4

Structure of a Synovial Joint

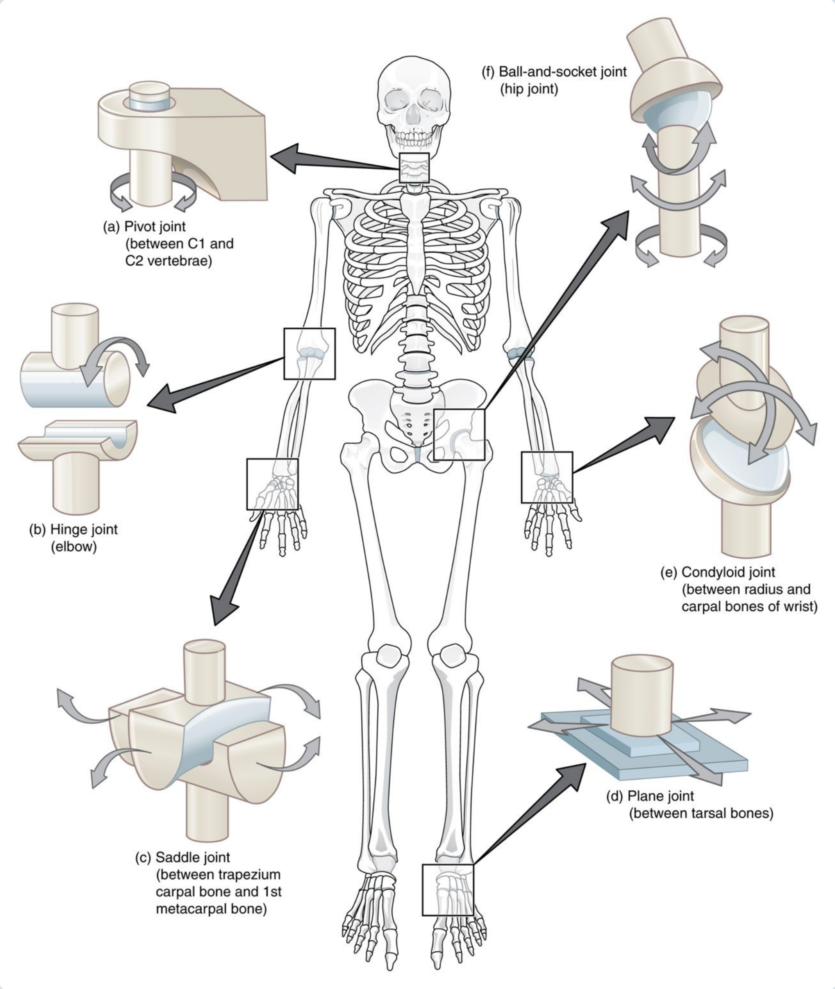


The structure of a synovial joint

- A fluid-filled joint cavity
- Articular cartilage caps the bone ends
- Synovial membrane secretes fluid
- Capsule and ligaments stabilize it

SECTION 9.4

Types of Synovial Joints



The six types of synovial joints

- Six shapes allow different motions
- Hinge (elbow), pivot (neck)
- Ball-and-socket (hip, shoulder)
- Saddle, plane, and condyloid

9.5

SECTION

Types of Body Movements

Flexion, extension, rotation, and the other movements joints allow.

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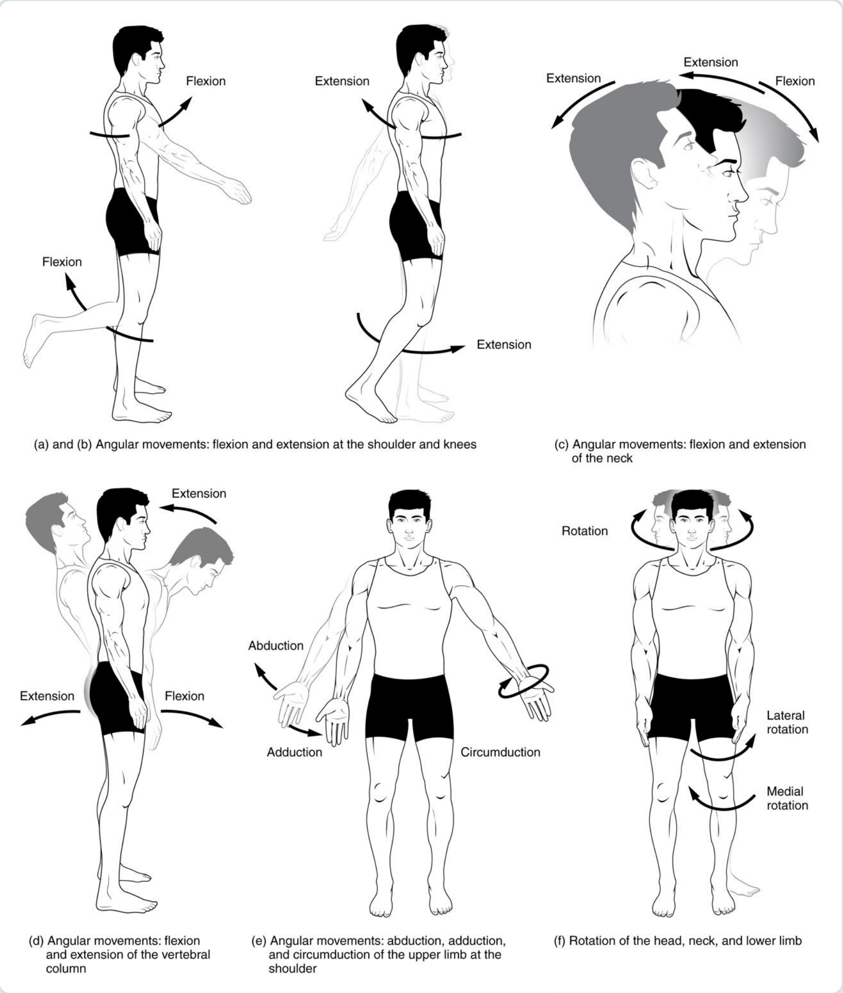
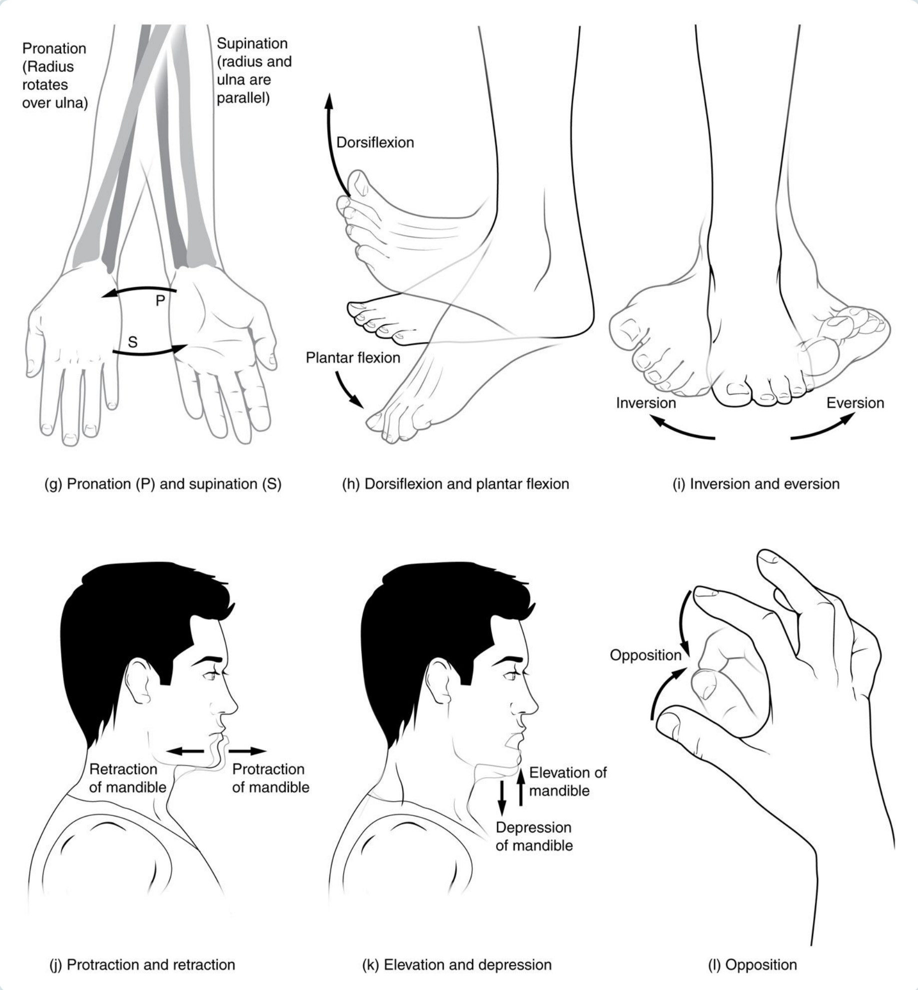
Body Movements — Angular



Angular movements at synovial joints

- Flexion decreases a joint angle
- Extension increases it
- Abduction moves away from midline
- Adduction moves toward it

Body Movements — Rotational & Special



Rotational and special movements

- Rotation turns a bone on its axis
- Pronation and supination of the forearm
- Inversion and eversion of the foot
- Special movements at specific joints

9.6

SECTION

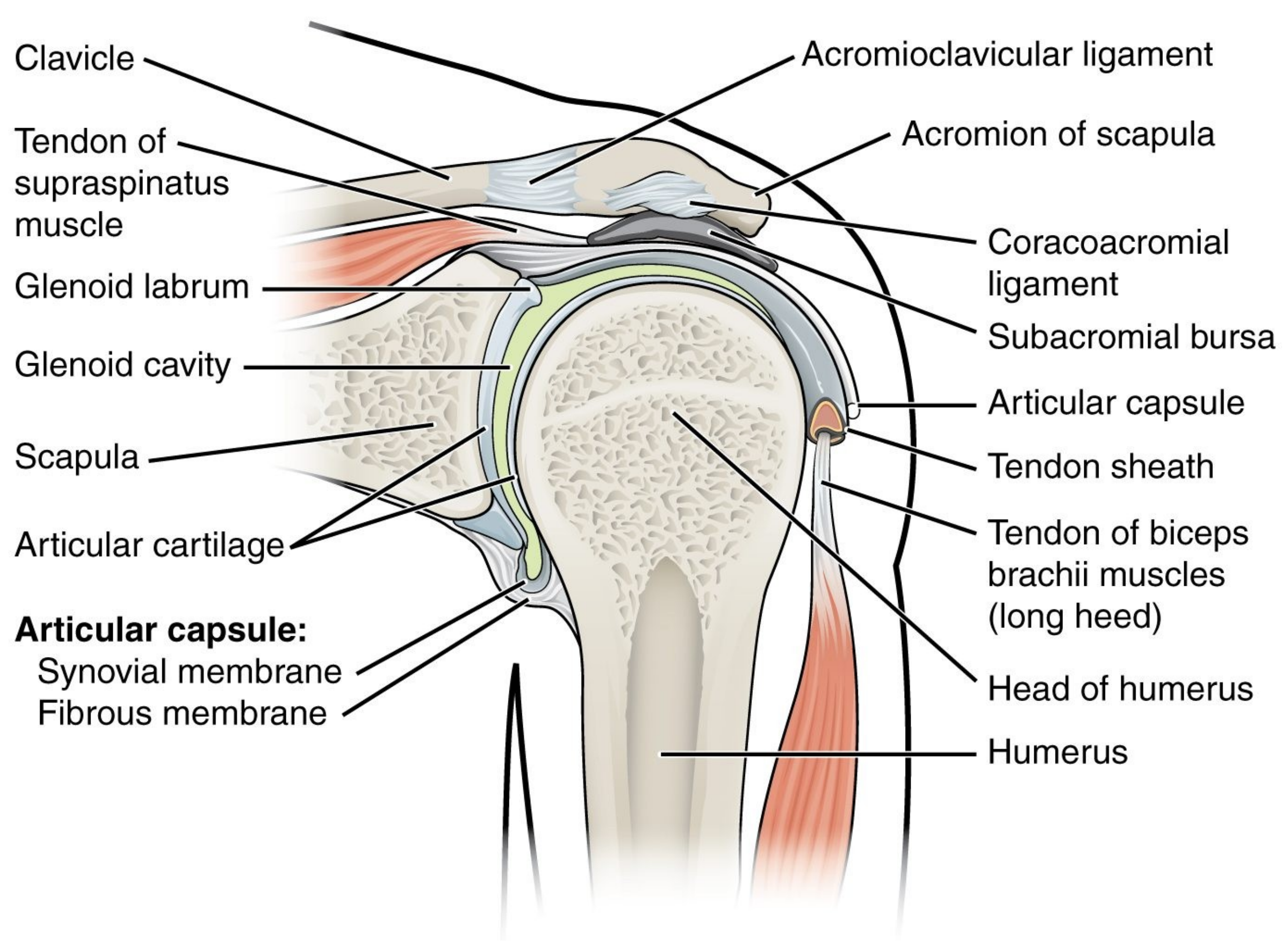
Anatomy of Selected Synovial Joints

A closer look at the shoulder, hip, and knee.

SECTION 9.6

The Shoulder Joint

- A ball-and-socket joint
- The most mobile joint in the body
- Shallow socket = least stable
- Rotator cuff provides support



The shoulder (glenohumeral) joint

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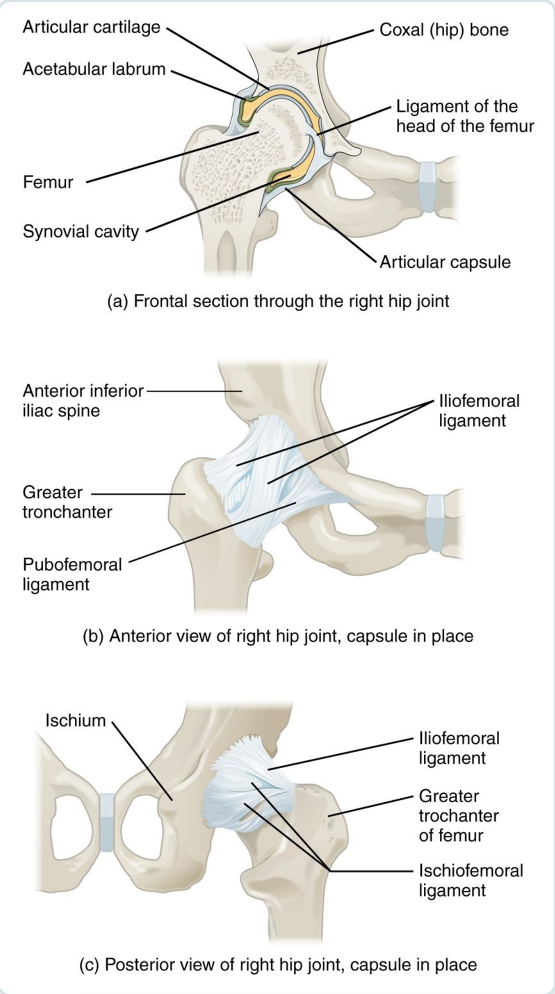
The Hip Joint

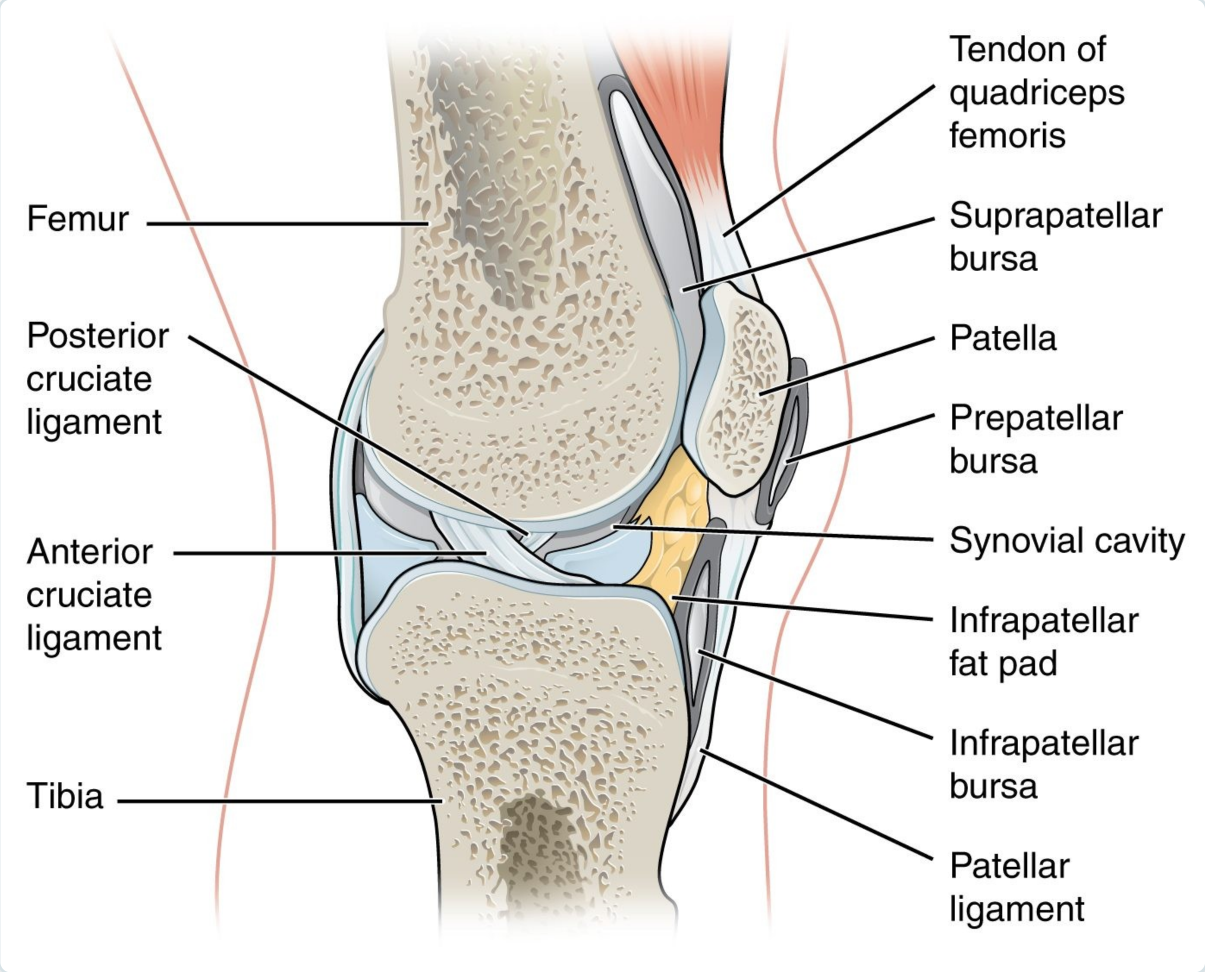


The hip joint

- A ball-and-socket joint
- Deep socket = very stable
- Bears the body's weight
- Strong ligaments reinforce it

The Knee Joint

- The largest, most complex joint
- A modified hinge joint
- Cruciate ligaments provide stability
- Menisci cushion the joint



The knee joint

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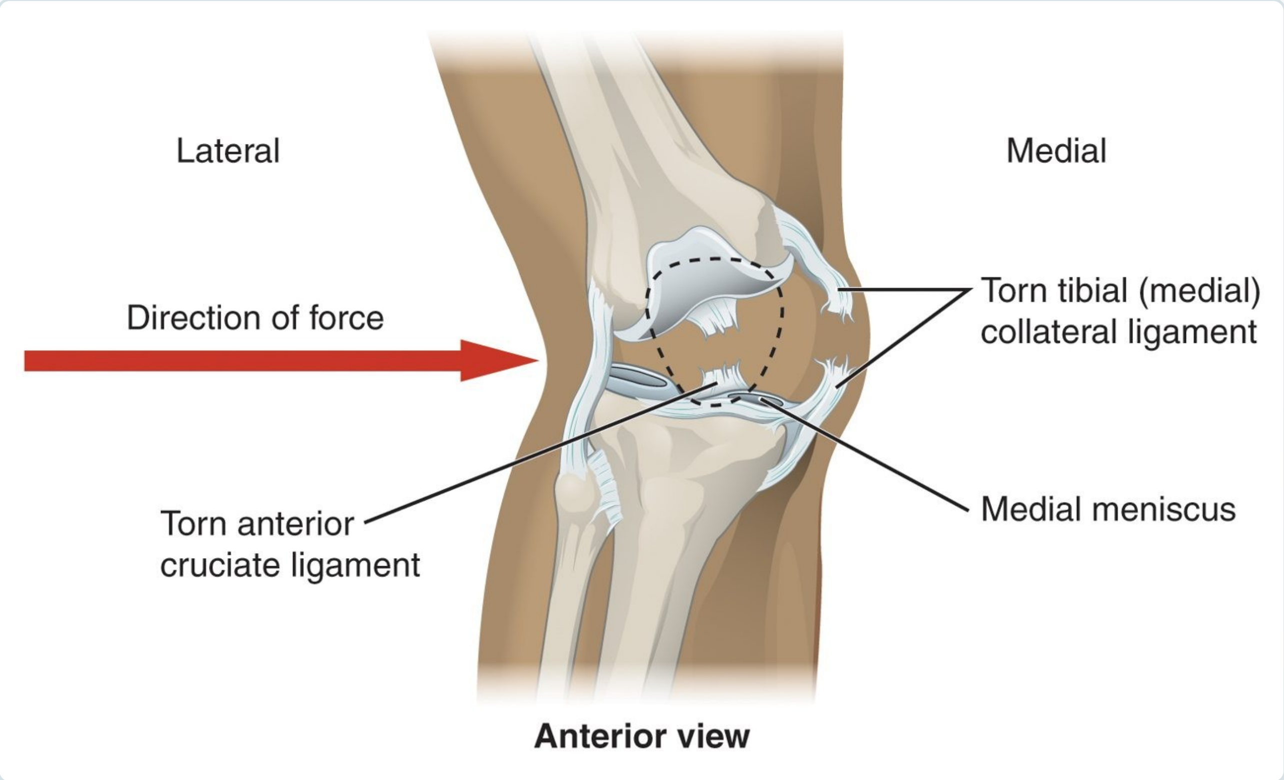
Knee Injuries



Torn ACL and meniscus of the knee

- The knee is prone to injury
- ACL tears from twisting forces
- Meniscus tears with rotation
- Common in sports and falls

9.7

SECTION

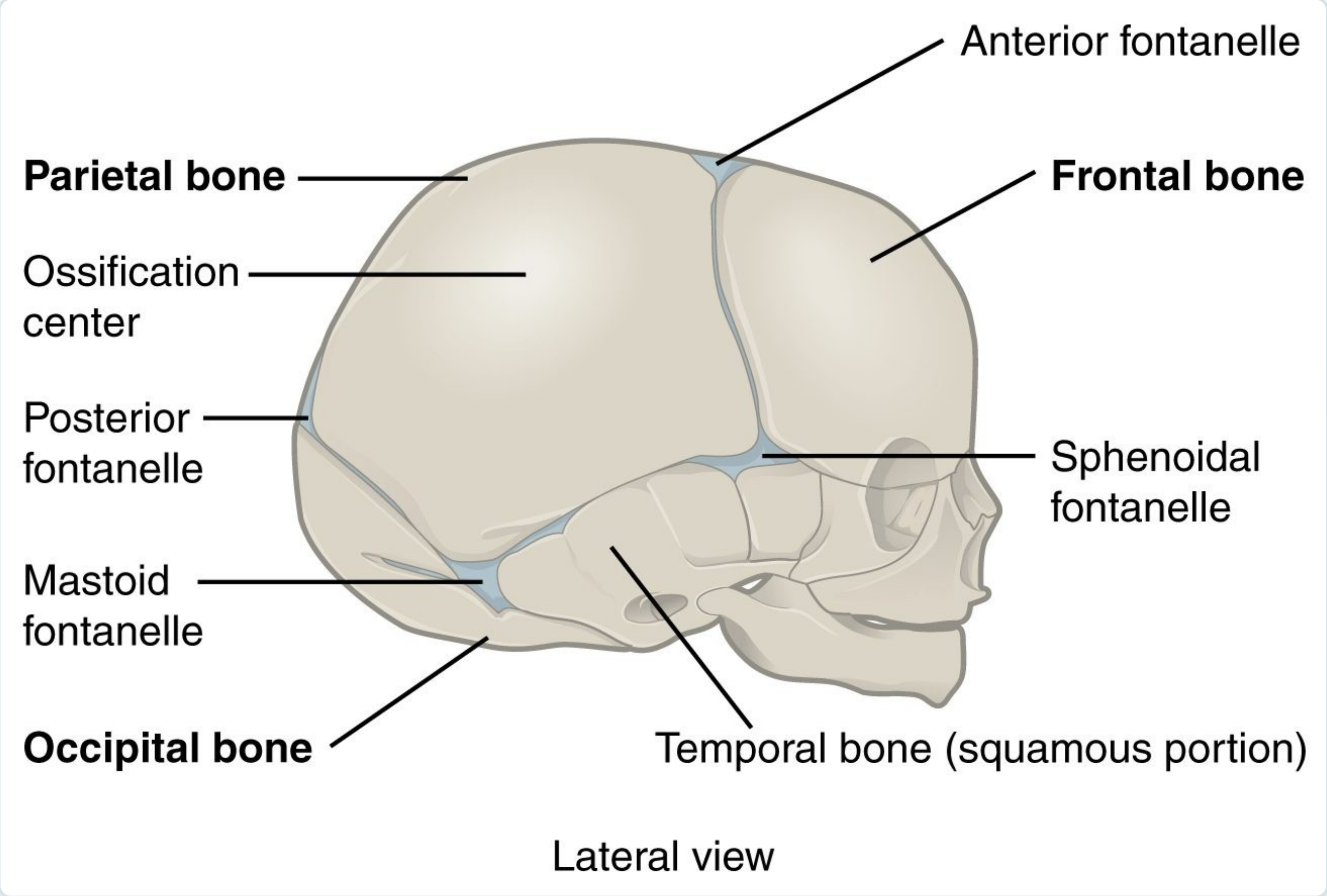
Development of Joints

How joints form before birth, as the skeleton develops.

SECTION 9.7

Development of Joints

- Joints form as the skeleton develops
- Fibrous fontanelles in the infant skull
- Allow skull molding and brain growth
- Close into sutures over time



Fontanelles of the developing skull

Key Takeaways

- Joints are classified by structure (fibrous, cartilaginous, synovial) and by how much they move.
- Fibrous and cartilaginous joints allow little movement but provide strength and shock absorption.
- Synovial joints are freely movable, with a fluid-filled cavity and six structural types.
- Body movements — flexion, extension, rotation, and more — describe how joints move.
- The shoulder, hip, and knee illustrate the trade-off between mobility and stability.

CLINICAL CONNECTIONS

Why Joints Matter in Practical Nursing

Range of Motion

Movement terms describe and document joint function and limitations.

Arthritis

Cartilage breakdown causes pain and stiffness — a leading cause of disability.

Joint Injuries

Sprains, dislocations, and ACL or meniscus tears are common.

Joint Replacement

Worn hips and knees are often replaced surgically.

Rehabilitation

Restoring joint movement is central to recovery and mobility.

Synovial Fluid

Changes in joint fluid can signal infection or inflammation.